
PROJECT TEAM

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Fruit Segmentation and Classification using GLCM Texture Analysis and DBSCAN Clustering

*A texture-driven approach to fruit recognition using own-code GLCM, DBSCAN,
and morphological refinement*

Problem Description

PROBLEM STATEMENT

Identify and segment different fruit classes using texture-based image analysis. Unlike conventional colour-based methods, this project relies primarily on texture information extracted from fruit surfaces.

ASSIGNMENT REQUIREMENTS

- **GLCM** — Gray-Level Co-occurrence Matrix for texture analysis
- **Autocorrelation** — Core GLCM-derived features
- **Custom DBSCAN** — Own-code density-based clustering algorithm
- **Morphology** — Opening and Closing operations for mask cleanup

KEY CHALLENGES

- ◆ Similar fruits share overlapping texture patterns
- ◆ Lighting variations affect texture measurements
- ◆ Real-world images contain noise & background clutter
- ◆ Segmentation must accurately separate fruit regions

EXPECTED OUTCOME

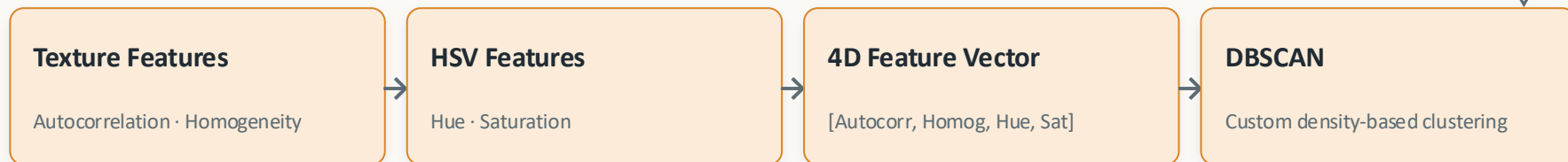
- ✓ **Accurate fruit segmentation**
- ✓ **Reliable fruit classification**
- ✓ **Validated on Fruits-360 Test set**
- ✓ **Evaluated on Fruits-262 real-world data**

Proposed Segmentation Workflow

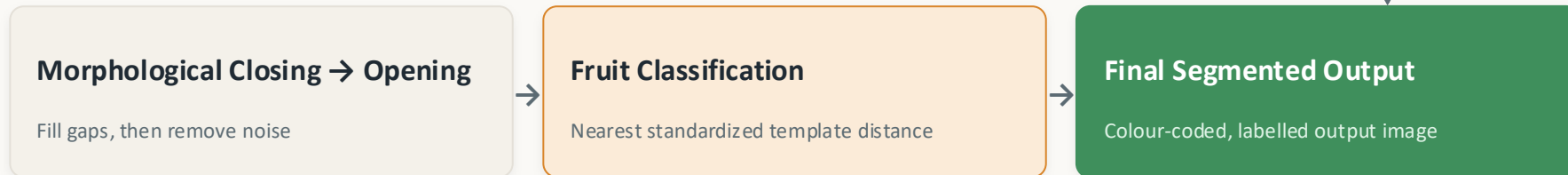
STAGE 1 — TEXTURE FEATURE EXTRACTION (GLCM)



STAGE 2 — FEATURE VECTOR & CLUSTERING



STAGE 3 — REFINEMENT & CLASSIFICATION



How the Algorithm Performs Segmentation

Understanding how each component contributes to separating fruit classes.

1. GLCM — TURNING TEXTURE INTO NUMBERS

- **The Mechanism:** A sliding 32x32 window scans the image, quantifying both physical roughness and light wavelength.
- **The Mathematics:** Extracts a unified 4-dimensional vector (Autocorrelation, Homogeneity, Hue, Saturation) for every local patch.
- **The Objective:** Maximizes feature orthogonality. Fruits with identical textures (e.g., Apple and Lemon) are mathematically forced apart by color, while identically colored fruits (e.g., Green Apple and unripe Pineapple) are separated by texture.

3. MORPHOLOGY

- **Closing Operation :** Seals internal gaps within the cluster using an elliptical kernel, ensuring that harsh surface glares or deep shadows do not mathematically fracture a single fruit into multiple pieces..
- **Opening Operation:** Strips away stray background artifacts and smooths the outer perimeter, forcing the final segmented mask to strictly adhere to the physical edge of the fruit.

2. DBSCAN — GROUPING WINDOWS INTO FRUITS

- **The Mechanism:** Projects the 4D vectors into a standardized spatial graph and searches for tightly packed data concentrations.
- **The Mathematics:** Clusters are formed organically based on proximity (epsilon) and density (min_samples), completely ignoring sparse background noise.
- **The Objective:** Eliminates the need to pre-define the number of targets. The algorithm dynamically discovers whether the environment contains 1, 3, or 10 fruits.

SEGMENTING LOGIC

- Same fruit → similar texture → High Spatial Density → Forms one cluster
- Different fruits → Orthogonal texture → Low proximity → Forms separate clusters
- Background / shadow → sparse → Labelled -1 (noise), Discarded as Noise

Fruit Class Selection — Texture-Based Reasoning

From the Fruits-360 training set, 10 classes were selected to span the full range of GLCM texture

SELECTION CRITERIA

- **Cover the texture spectrum** — from very smooth (Green Apple) to highly granular (raspberry)
- **Distinct GLCM signatures** — minimize overlap in autocorrelation / homogeneity

EXPECTED CLUSTER SEPARATION

Pairwise risk of cluster overlap in standardized 4D feature space, based on GLCM + Hue/Saturation similarity:

Low risk pairs: Raspberry vs Green Apple, Avocado vs Lemon — opposite ends of the texture spectrum

Moderate risk pairs: Cantaloupe vs Salak, Pineapple vs Kiwi — similar roughness, different colour

Watch-list pair: Strawberry vs Cantaloupe — both moderately granular; separated mainly by hue/saturation

SELECTED CLASSES (TOP PRIORITY)

Raspberry

Pineapple

Cantaloupe

OTHER CANDIDATE CLASSES

Lemon

Kiwi

Strawberry

Avocado

Watermelon

Salak

Green Apple

Key Code — Training & Segmentation

TEMPLATE BUILDER (GLCM BOUNDARIES)

Builds per-fruit GLCM + HSV feature templates (mean, std) from labelled training images.

GLCM FEATURE EXTRACTION

```
glcm = np.zeros((256, 256), dtype=np.float64)
np.add.at(glcm, (left_pixels, right_pixels), 1)

total_pairs = np.sum(glcm)
if total_pairs > 0:
    glcm_norm = glcm / total_pairs
    i, j = np.indices((256, 256), dtype=np.float64)
    autocorr = np.sum(i * j * glcm_norm)
    homo = np.sum(glcm_norm / (1 + (i - j)**2))
```

TEMPLATE RANGE BUILDER

```
return {
    "autocorr_min": float(np.percentile(all_auto, 20)),
    "autocorr_max": float(np.percentile(all_auto, 80)),
    "homo_min": float(np.percentile(all_homo, 20)),
    "homo_max": float(np.percentile(all_homo, 80)),
    "hue_min": float(np.percentile(all_hue, 20)),
    "hue_max": float(np.percentile(all_hue, 80)),
    "sat_min": float(np.percentile(all_sat, 20)),
    "sat_max": float(np.percentile(all_sat, 80))
}
```

SEGMENTATION ENGINE (4D DBSCAN)

Applies the trained templates: extracts features, runs custom DBSCAN, classifies clusters, and renders the segmented output.

CUSTOM DBSCAN ALGORITHM

```
def oc_dbscan(features, eps, min_samples):
    n = features.shape[0]
    labels = -1 * np.ones(n, dtype=int)
    cluster_id = 0
    for i in range(n):
        if labels[i] != -1:
            continue
        distances = np.linalg.norm(features - features[i], axis=1)
        neighbors = np.where(distances <= eps)[0]
        if len(neighbors) < min_samples:
            labels[i] = -1
        else:
            labels[i] = cluster_id
            seed_set = list(neighbors)
            seed_set.remove(i)
            while len(seed_set) > 0:
                current_point = seed_set.pop(0)
                if labels[current_point] == -1:
                    labels[current_point] = cluster_id
                ...
            cluster_id += 1
    return labels
```

Quantitative Baseline Validation (3-Class)

OBJECTIVE

Validate the core mathematical logic (GLCM) in a controlled environment prior to real-world integration testing.

METHODOLOGY

Manual evaluation of 15 random samples (5 per class) from the controlled Test dataset using un-relaxed, studio-derived feature boundaries.

THE CONFUSION MATRIX

15 / 15 samples correctly clustered — perfect diagonal under controlled (studio) conditions

	PREDICTED Cantaloupe 2	PREDICTED Pineapple	PREDICTED Raspberry
ACTUAL Cantaloupe 2	5	0	0
ACTUAL Pineapple	0	5	0
ACTUAL Raspberry	0	0	5

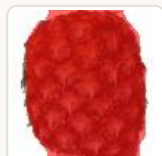
TEST LOGS (15/15 PASSED)

Cantaloupe 2: 8_100.jpg, 25_100.jpg, 98_100.jpg, r_28_100.jpg, r_35_100.jpg | **Pineapple:** 32_100.jpg, 65_100.jpg, 103_100.jpg, 229_100.jpg, 290_100.jpg
Raspberry: 12_100.jpg, 124_100.jpg, 129_100.jpg, 147_100.jpg, 267_100.jpg

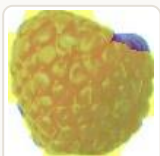
Dataset Sample Gallery — Fruits-360 vs Fruits-262

Representative test images from the two datasets used to evaluate the pipeline: Fruits-360 (clean studio crops) and Fruits-262 (real-world segmented samples).

FRUITS-360 — TRAINING CLASSES (STUDIO CROPS)



Pineapple



Raspberry



Cantaloupe



Lemon



Kiwi



Strawberry



Avocado



Watermelon



Salak



Green Apple

FRUITS-262 — REAL-WORLD SEGMENTED OUTPUTS



Cantaloupe (segmented)



Pineapple (segmented)



Raspberry (segmented)

Proof of Concept — Using Synthetic Data

Before evaluating on real photographs, the pipeline was validated on synthetically composed scenes (multiple Fruits-360 fruits placed together on a single canvas) — confirming that GLCM + DBSCAN can correctly separate and label distinct fruit regions within one image.

SYNTHETIC INPUT



Synthetic input image

CLASSIFIED OUTPUT



Segmented / classified output



Cantaloupe



Pineapple



Raspberry

Technical Parameters:

Resolution: 8k Macro.

Pipeline: 4D Feature Space (Auto, Homo, Hue, Sat).

DBSCAN Params: epsilon = 1.25, min_samples = 10, Squelch = 254

Best Ranking & Performing Classes

TOP 3 BEST-PERFORMING CLASSES

1 Raspberry

Highly granular, seed-rich texture produces a very distinctive GLCM signature (high contrast, low homogeneity) with minimal overlap with other classes — clusters form tightly and reliably.

2 Pineapple

Coarse, repetitive rind pattern gives strong, stable autocorrelation values across windows; its rough texture is visually and statistically far from smoother fruits.

3 Cantaloupe

Distinct net-like reticulated skin creates a unique mid-range homogeneity/contrast combination that rarely collides with other fruit templates.

WORST-PERFORMING CLASS

Lemon and Green Apple

Both classes lack a strong, distinctive surface texture — the primary signal this pipeline relies on — so they fall back on weaker hue/saturation cues that are themselves shared with neighboring fruit classes.

GENERAL INSIGHT

Classes with high-frequency, repetitive surface texture (seeds, netting, coarse rind) produce GLCM statistics that sit far from the global feature-space centroid — making their clusters easy for DBSCAN to isolate at a fixed eps.

Conversely, smooth, low-contrast, or visually “average” textures tend to sit near other classes in standardized 4D space, increasing the risk of cluster merging or misclassification.

References

[1]

Haralick, R. M., Shanmugam, K., & Dinstein, I. (1973). Textural Features for Image Classification. IEEE Transactions on Systems, Man, and Cybernetics, SMC-3(6), 610–621.

GLCM — Autocorrelation & Homogeneity features

[2]

Ester, M., Kriegel, H. P., Sander, J., & Xu, X. (1996). A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise. Proceedings of KDD-96 (pp. 226–231). AAAI Press.

DBSCAN clustering algorithm

[3]

Gonzalez, R. C., & Woods, R. E. (2018). Digital Image Processing (4th ed.). Pearson. (Chapter 9: Morphological Image Processing).

Mathematical morphology — Opening & Closing

[4]

<https://davinci.ai/app>

Prompt: A photorealistic, highly detailed image of a whole cantaloupe melon, a whole spiky pineapple, and a handful of whole fresh red raspberries sitting together in a simple white ceramic bowl. The bowl is on a plain white studio table with bright, even, diffused lighting. No harsh shadows. 8k resolution, macro photography